Chapter 3: Literature Survey

A thorough review of existing literature and systems was conducted to understand the current landscape of agricultural waste management and related technologies.

3.1 Agricultural Waste Management in India

Jain et al. (2014) estimated that approximately 92 million tonnes of crop residue are burned annually in India, contributing significantly to air pollution. Their study highlighted the need for technology-driven alternatives to open burning. The Indian Agricultural Research Institute (IARI) has documented the environmental and health impacts of stubble burning across multiple studies.

3.2 Digital Platforms for Waste Management

Several digital platforms have emerged for general waste management. Platforms like "Swachhata" by the Ministry of Housing focus on urban waste, while "mKisan" provides agricultural advisory services. However, none specifically address the agricultural waste tracking and recycling workflow that AgriTrace targets. The gap between general-purpose waste management apps and agricultural-specific requirements motivated the development of AgriTrace.

3.3 Firebase and Real-Time Database Systems

Google Firebase provides a comprehensive suite of backend services including Firestore (NoSQL database), Authentication, and Cloud Storage. Studies by Moroney (2017) have demonstrated Firebase's effectiveness in building real-time applications with offline support. The security rules system provides granular access control, which is critical for role-based applications like AgriTrace.

3.4 Next.js and Modern Web Frameworks

Next.js, developed by Vercel, has become the leading React framework for production applications. Version 15.5 introduces Turbopack for faster development builds and the App Router for improved routing and server components. TypeScript integration ensures type safety across the entire codebase.

3.5 Carbon Credit Systems

The carbon credit market has evolved significantly with the Paris Agreement. Studies by Pandey and Agrawal (2014) have quantified the carbon emissions from crop residue burning in India. AgriTrace uses scientifically established emission factors to calculate carbon offsets, with values ranging from 980 to 1400 kg CO_2 per metric tonne depending on waste type.

3.6 Payment Gateway Integration

Razorpay has emerged as India's leading payment gateway, supporting UPI, cards, and net banking. Its well-documented API and webhook system make it suitable for B2B and B2C transaction processing in agricultural marketplaces. Razorpay provides test mode capabilities, allowing developers to simulate payment flows without processing real transactions. The Orders API creates a server-side order with a specified amount, and the checkout.js library renders a user-facing payment modal that handles card validation, OTP verification, and payment confirmation. Upon successful payment, Razorpay generates a signature that must be verified server-side to prevent tampering.

3.7 IoT and Sensor-Based Waste Monitoring

Research by Kumar et al. (2020) explored IoT-based monitoring of agricultural waste burning using satellite imagery and ground-based sensors. Their work demonstrated that real-time monitoring can significantly improve enforcement of burning regulations. While AgriTrace does not directly integrate IoT sensors, its GPS-based location tracking and photo verification features serve a similar purpose by providing digital evidence of waste collection activities. Future versions of AgriTrace could incorporate IoT sensor data to automate waste quantity measurement and quality assessment.

3.8 Mobile-First Web Applications for Rural India

Studies by Patel et al. (2019) have highlighted the importance of mobile-first design for applications targeting rural Indian users. With smartphone penetration increasing rapidly in rural areas, web applications must be responsive, lightweight, and functional under intermittent network conditions. AgriTrace addresses this through responsive design using Tailwind CSS breakpoints, optimistic UI updates with Firebase's offline persistence, and progressive image loading for waste photographs. The application is designed to function meaningfully even on devices with limited bandwidth.

3.9 Tailwind CSS and Utility-First Design Systems

The utility-first CSS methodology, popularized by Tailwind CSS (Wathan, 2017), represents a paradigm shift from traditional CSS approaches. Rather than writing custom CSS classes, developers compose styles using atomic utility classes directly in HTML markup. Research by State of CSS Survey (2023) found that Tailwind CSS adoption has grown to over 50% among professional developers, citing improved development speed, consistency, and reduced CSS bundle sizes as key benefits. AgriTrace leverages Tailwind's responsive modifiers (e.g., `sm:`, `md:`, `lg:`), dark mode support, and arbitrary value syntax to create a polished, consistent user interface.

3.10 Comparative Analysis of Existing Agricultural Platforms

The following table presents a comparative analysis of existing agricultural platforms and their limitations compared to AgriTrace:

Table 3.1: Comparative Analysis of Existing Agricultural Platforms

Feature	**Kisan Suvidha**	**mKisan**	**Swachhata**	**PUSA Decomposer**	**AgriTrace**
Waste Tracking	No	No	Partial	No	Yes
Collection Coordination	No	No	No	No	Yes
Payment Integration	No	No	No	No	Yes
Carbon Credits	No	No	No	No	Yes
Real-time Updates	No	Partial	Partial	No	Yes
Role-Based Access	No	No	Yes	No	Yes
GPS Location	Partial	No	Yes	No	Yes
Photo Verification	No	No	Yes	No	Yes

As the table demonstrates, AgriTrace addresses a unique combination of features that no existing platform currently provides, making it a novel contribution to the agricultural waste management domain.

3.11 Summary of Literature Review

From the literature survey, it is evident that while considerable research exists on agricultural waste management, digital platforms for waste management, and modern web technologies, there is a significant gap in integrated solutions that combine waste tracking, marketplace functionality, payment processing, carbon credit quantification, and gamification. AgriTrace is positioned to fill this gap by leveraging established technologies (Next.js, Firebase, Razorpay) in a novel application domain. The literature review also confirms the choice of Agile methodology, NoSQL databases, and responsive design as appropriate architectural decisions for this type of application.